

练习四 力学基本定律（四）

1. $\sqrt{3gl}$

2. $0.6\omega_0; \frac{5\omega_0}{4\pi}$

3. (1);

4. (3);

5. $m_1g - T_1 = m_1a_1 \quad T_2 - m_2g = m_2a_2 \quad T_1R - T_2r = (J_1 + J_2)\alpha$

$$a_1 = R\alpha \quad a_2 = r\alpha$$

联立解得: $\alpha = \frac{(m_1R - m_2r)g}{J_1 + J_2 + m_1R^2 + m_2r^2}$

$$a_1 = \frac{(m_1R - m_2r)Rg}{J_1 + J_2 + m_1R^2 + m_2r^2}$$

$$a_2 = \frac{(m_1R - m_2r)rg}{J_1 + J_2 + m_1R^2 + m_2r^2}$$

$$T_1 = \frac{J_1 + J_2 + m_2r(R+r)}{J_1 + J_2 + m_1R^2 + m_2r^2} m_1g$$

$$T_2 = \frac{J_1 + J_2 + m_1R(R+r)}{J_1 + J_2 + m_1R^2 + m_2r^2} m_2g$$

6. $\frac{1}{2} \cdot \frac{1}{3} ml^2 \omega_0^2 = mg \frac{l}{2} \quad \omega_0 = \sqrt{\frac{3g}{l}}$

$$\frac{1}{2} \cdot \frac{1}{3} ml^2 \omega_0^2 = \frac{1}{2} \cdot \frac{1}{3} ml^2 \omega^2 + \frac{1}{2} mv^2$$

$$\frac{1}{3} ml^2 \omega_0 = \frac{1}{3} ml^2 \omega + lmv$$

$$v = \frac{1}{2} \sqrt{3gl}$$